ChainOps
SMART CONTRACT OPERATIONS

# ChainOps

Smart Contract DevSecOps and
Operations Platform






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# Meet Our Team



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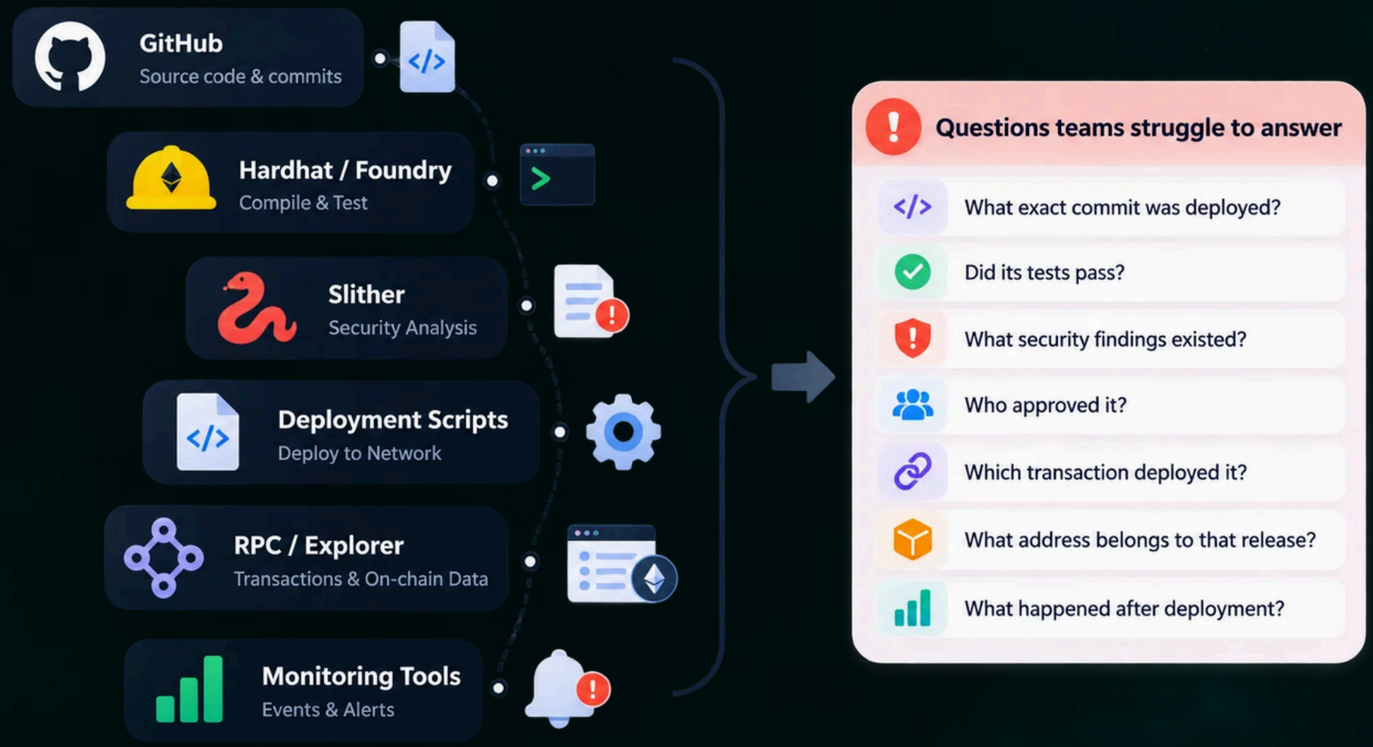


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# PROJECT OVERVIEW

## The Problem

Smart-contract delivery is fragmented across multiple tools



# PROJECT OVERVIEW

## Our Solution

ChainOps creates one traceable smart-contract lifecycle




ChainOps does not replace GitHub, Hardhat, Slither or blockchain infrastructure. It orchestrates them and adds lifecycle traceability, governance and operational visibility

# AIM AND OBJECTIVE

## Our Aim

Build a Smart Contract DevSecOps and operations platform connecting GitHub development with automated validation, security governance, controlled deployment and post-deployment monitoring.



# AIM AND OBJECTIVE

## Our Objectives

BUILD

- Automate source revision checkout, compilation and testing

SECURE

- Perform static analysis and evaluate release policies

GOVERN

- Control sensitive deployments through approval workflows

DEPLOY

- Track EVM transactions and maintain a contract registry

OBSERVE

- Monitor deployed contracts and generate operational alerts

# SYSTEM SCOPE

## What ChainOps includes and what is left out

### IN SCOPE

- EVM-compatible Solidity projects
- GitHub integration
- Automated pipelines
- Build & testing
- Security analysis
- Policy & approvals
- Testnet deployment
- Contract registry
- Monitoring & alerts
- Audit & notifications

### OUT OF SCOPE

- New blockchain
- New compiler
- New vulnerability analyzer
- Real-mainnet funds
- Every blockchain ecosystem
- Autonomous AI deployments
- Generic CI/CD for all software

# TARGET USERS

## Who uses and interacts with ChainOps




Solo development is also supported, where one developer can operate a project without requiring a formal organization structure.

# KEY MODULES & FUNCTIONS

## Six integrated areas powering the ChainOps lifecycle

### Platform & Access
Auth • Projects • RBAC

### Pipeline & Execution
GitHub • Build/Test • Runs

### Security & Governance
Findings • Policies • Approvals



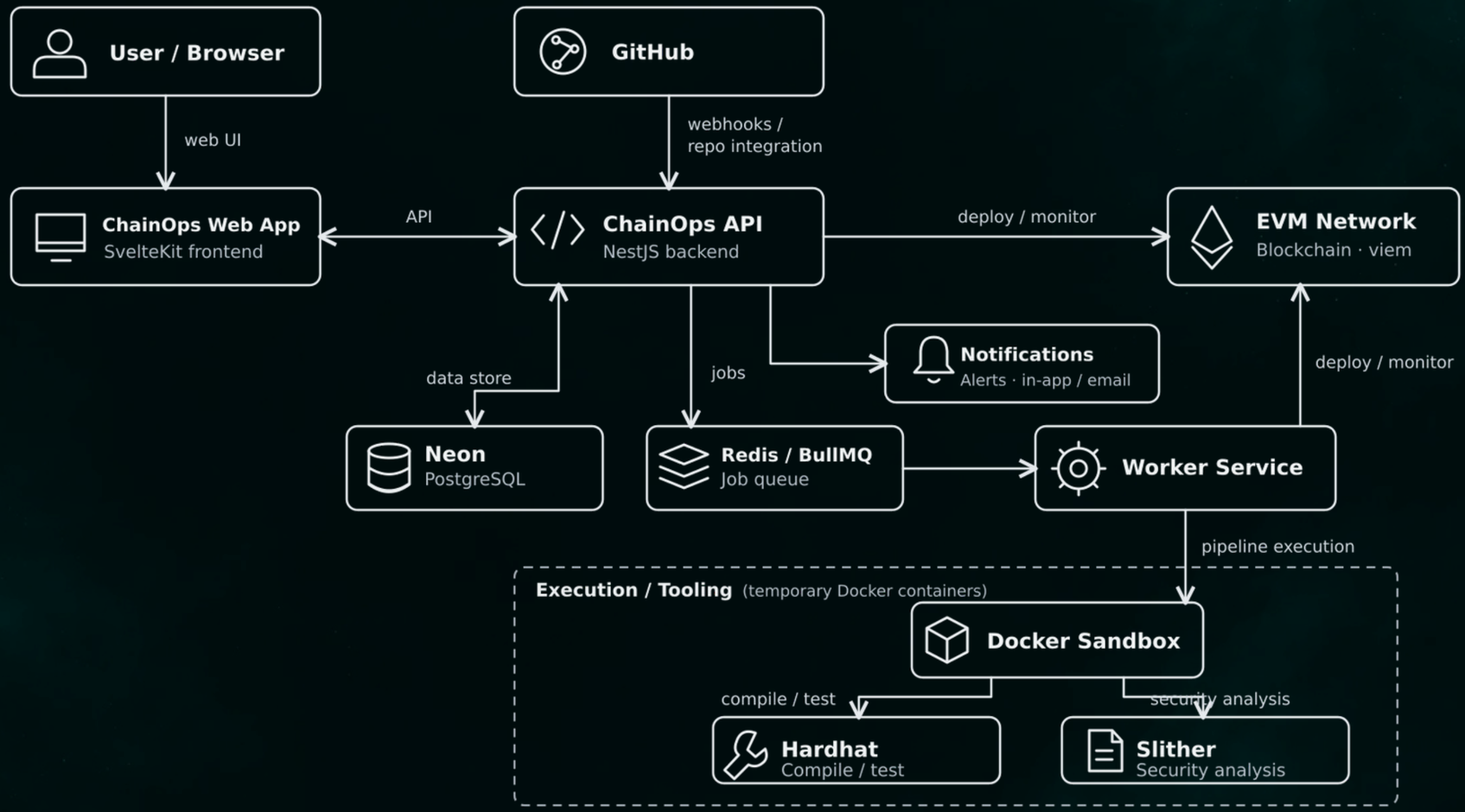

### Deployment & Registry
EVM Deploy • Transactions • Registry

### Monitoring & Alerts
Events • Rules • Notifications

### Audit & AI
Audit Trail • Reports • Assistant

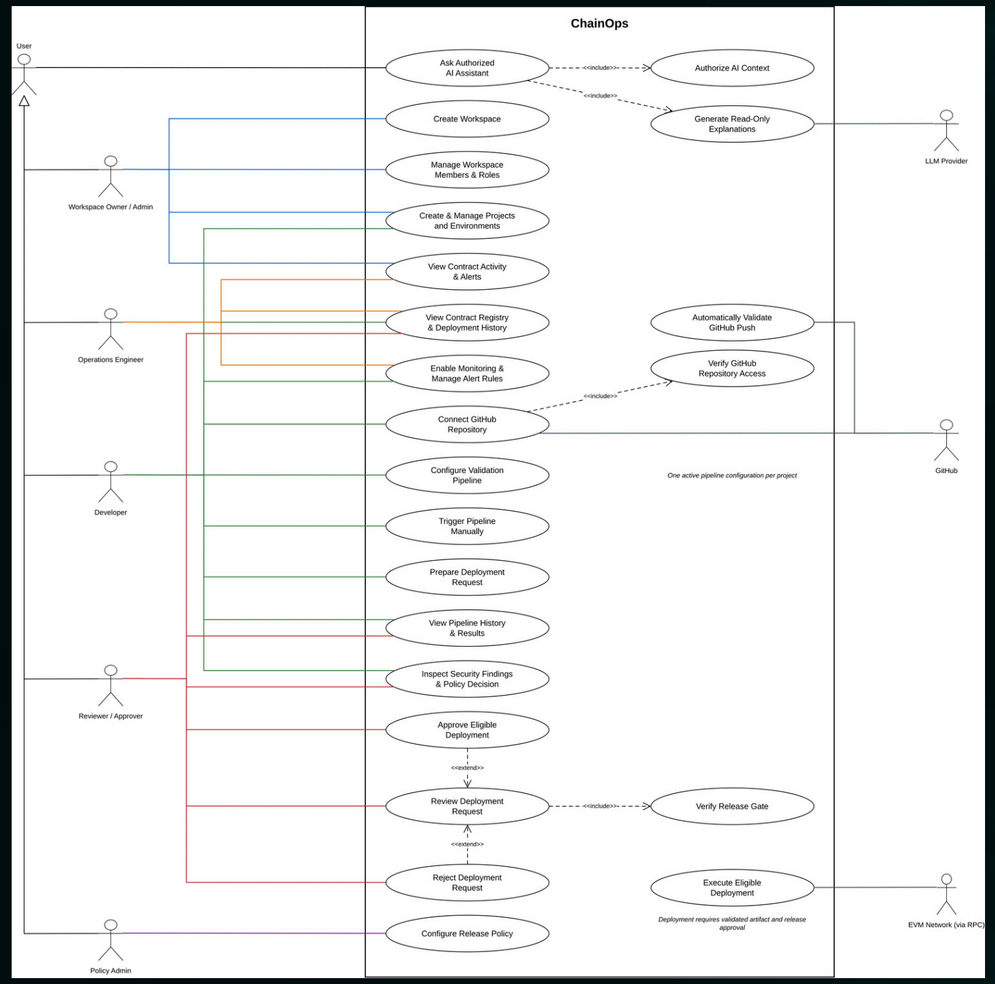
# SYSTEM DIAGRAMS

## System Architecture



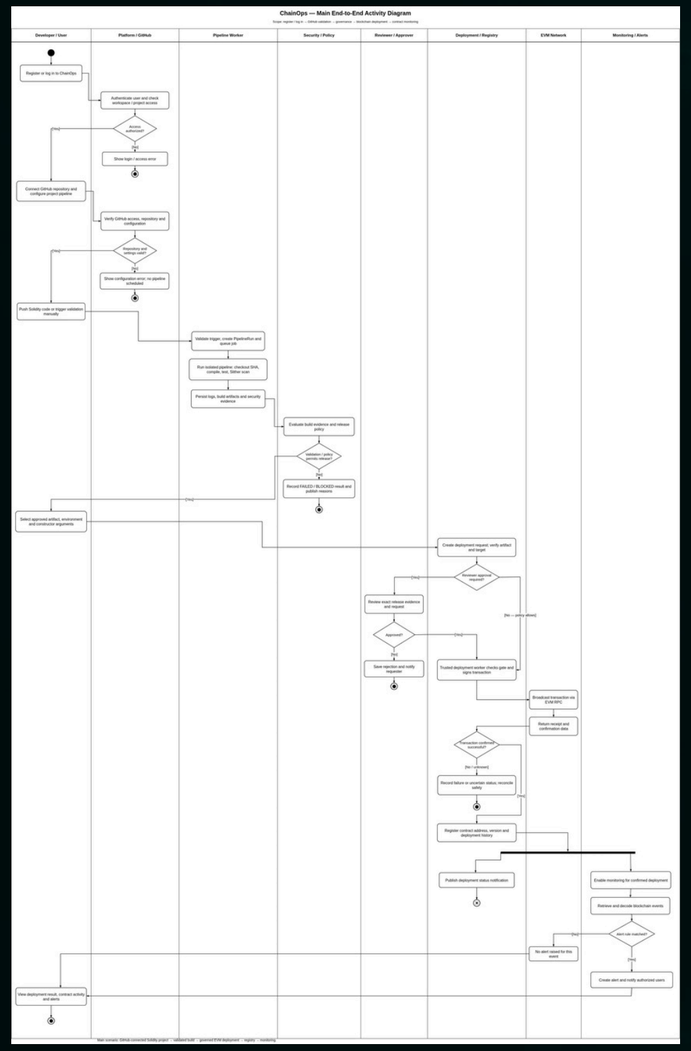
# SYSTEM DIAGRAMS

## Use-Case Diagram



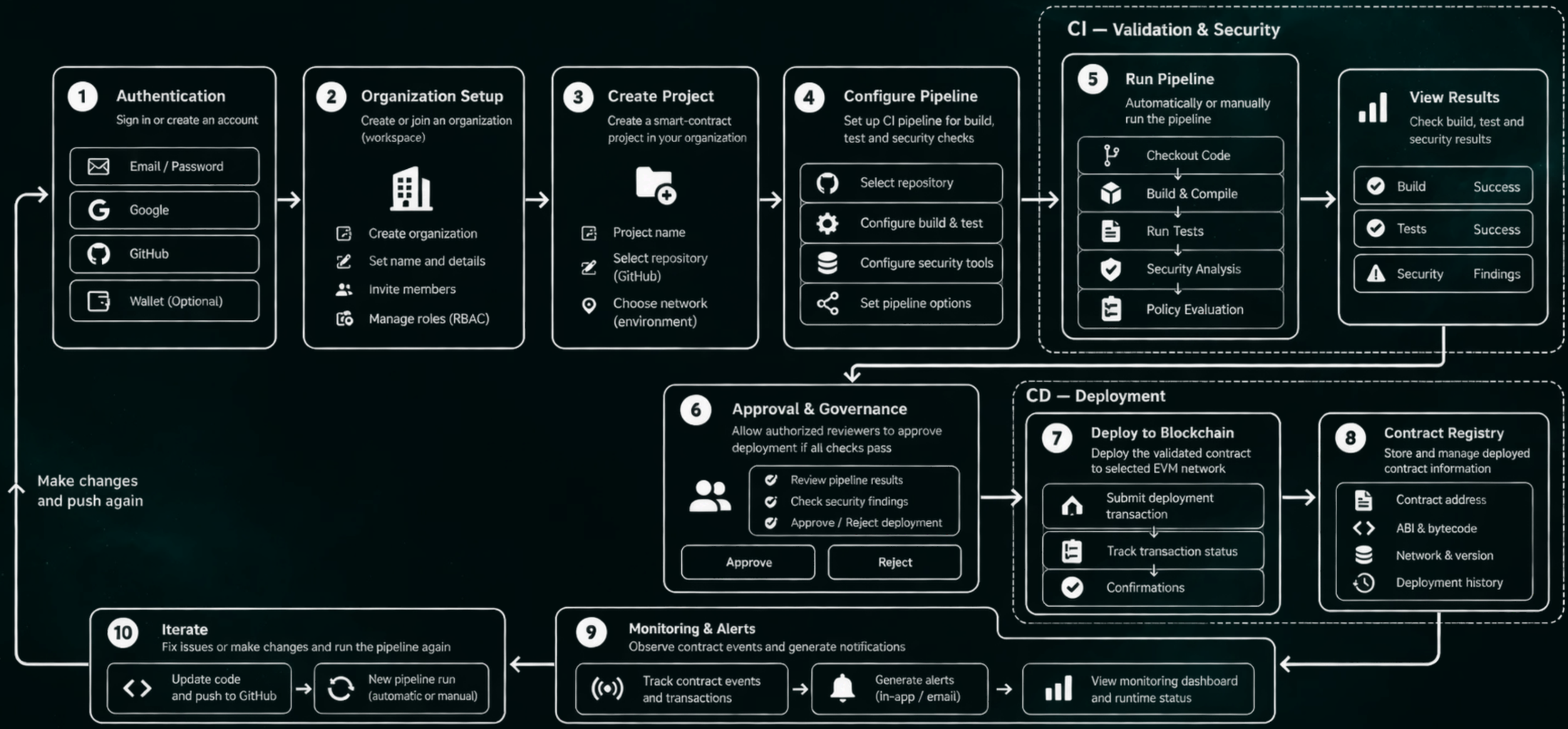
# SYSTEM DIAGRAMS

## Activity Diagram



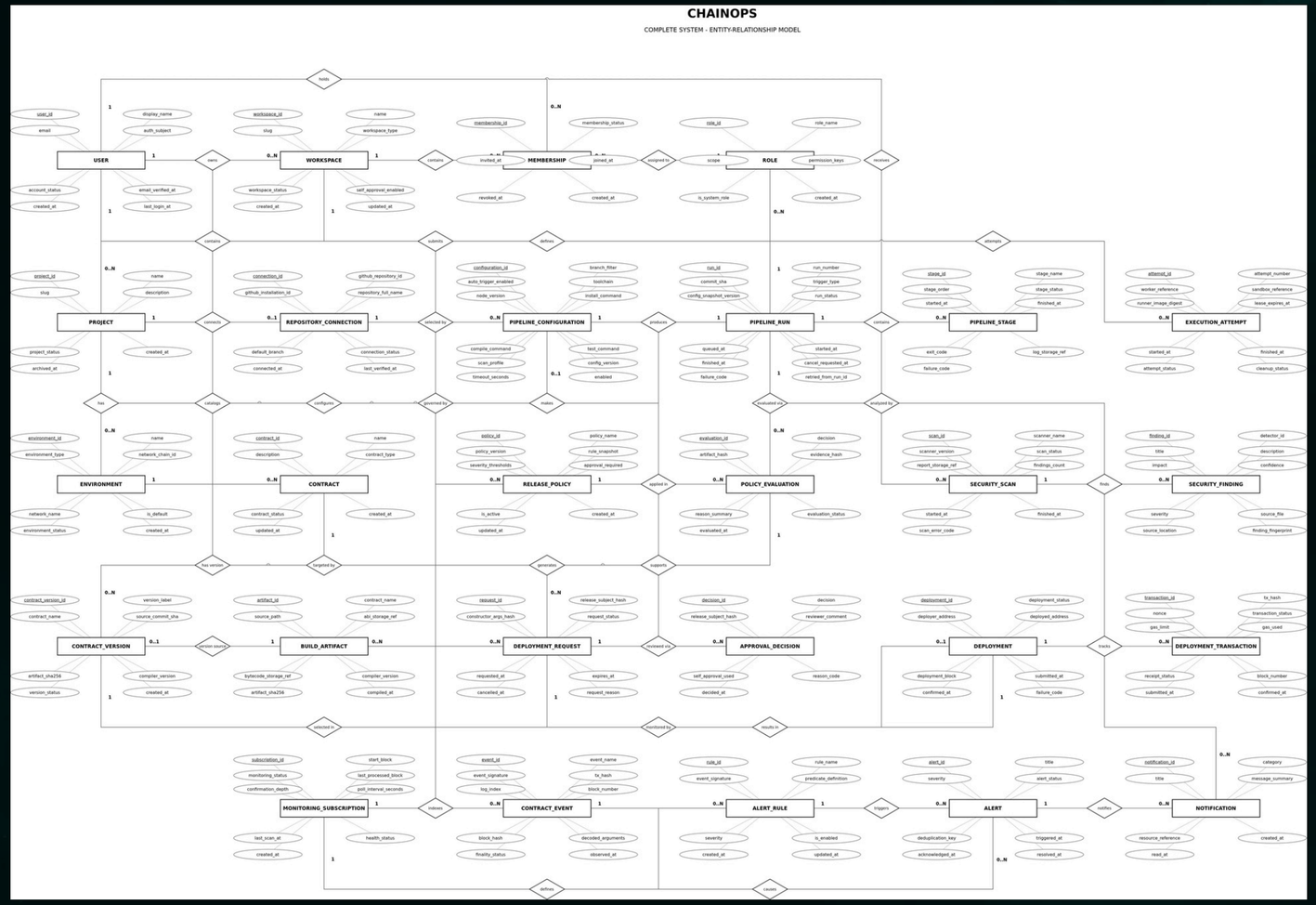
# SYSTEM DIAGRAMS

## System Flow



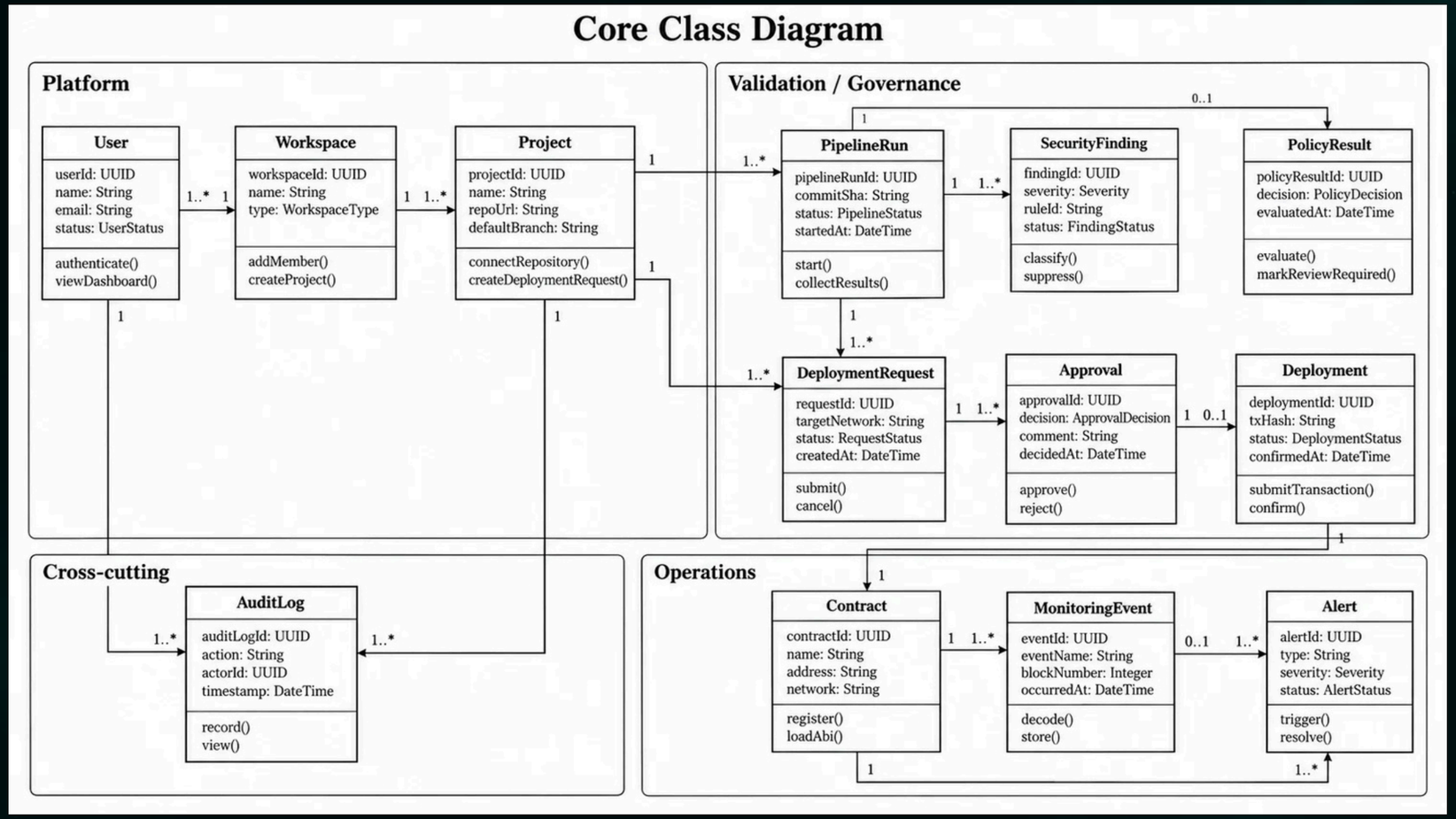
# SYSTEM DIAGRAMS

## ER Diagram



# SYSTEM DIAGRAMS

## Class Diagram



# TECH STACK

## Core Platforms and tools used in ChainOps

### Frontend


React



TypeScript

### Backend & Data


NestJS


Prisma


PostgreSQL


Neon


Redis


BullMQ

### Blockchain & Security


Solidity


Hardhat



viem


Slither

### DevOps & Design


Docker

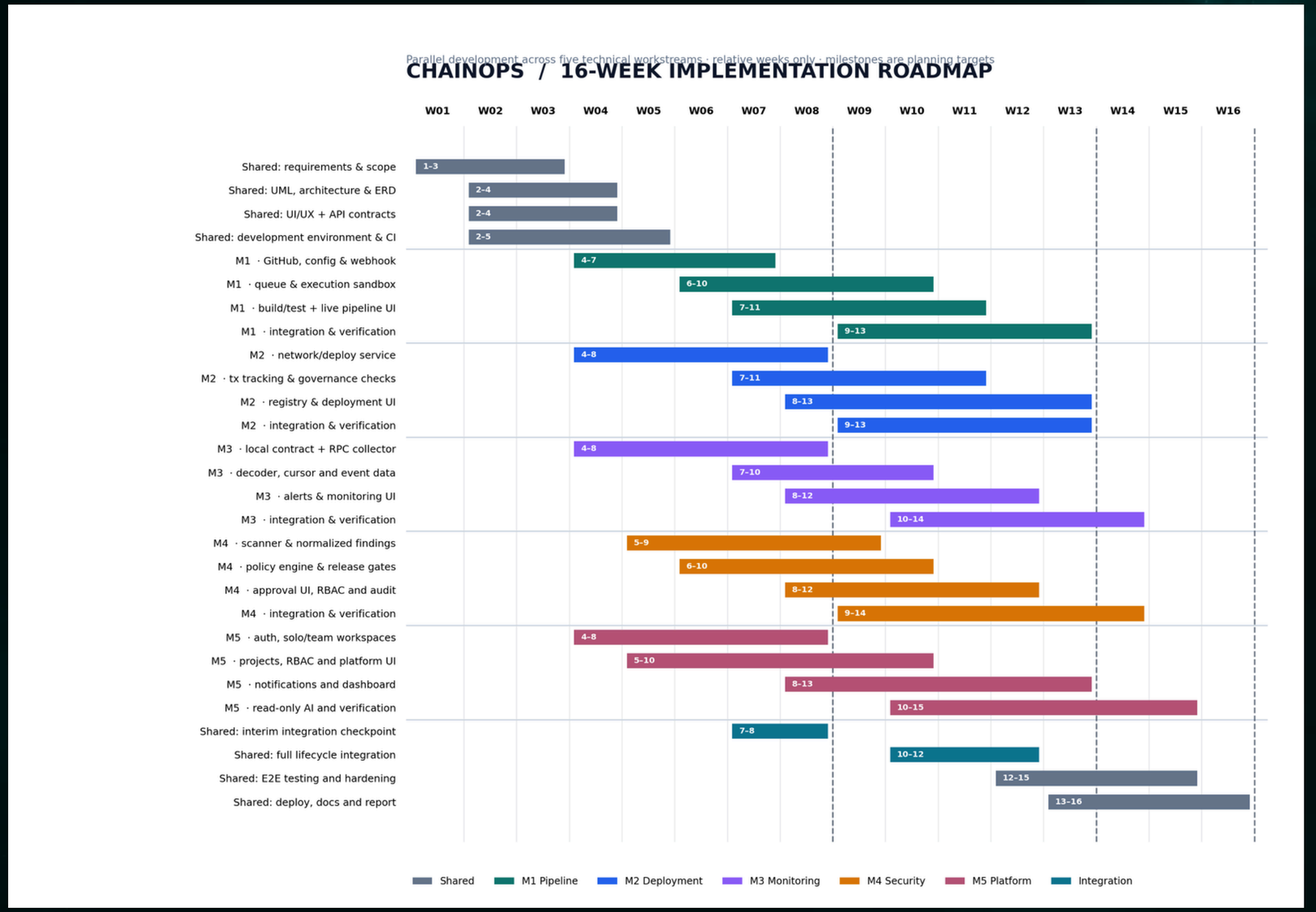

GitHub


GitHub Actions

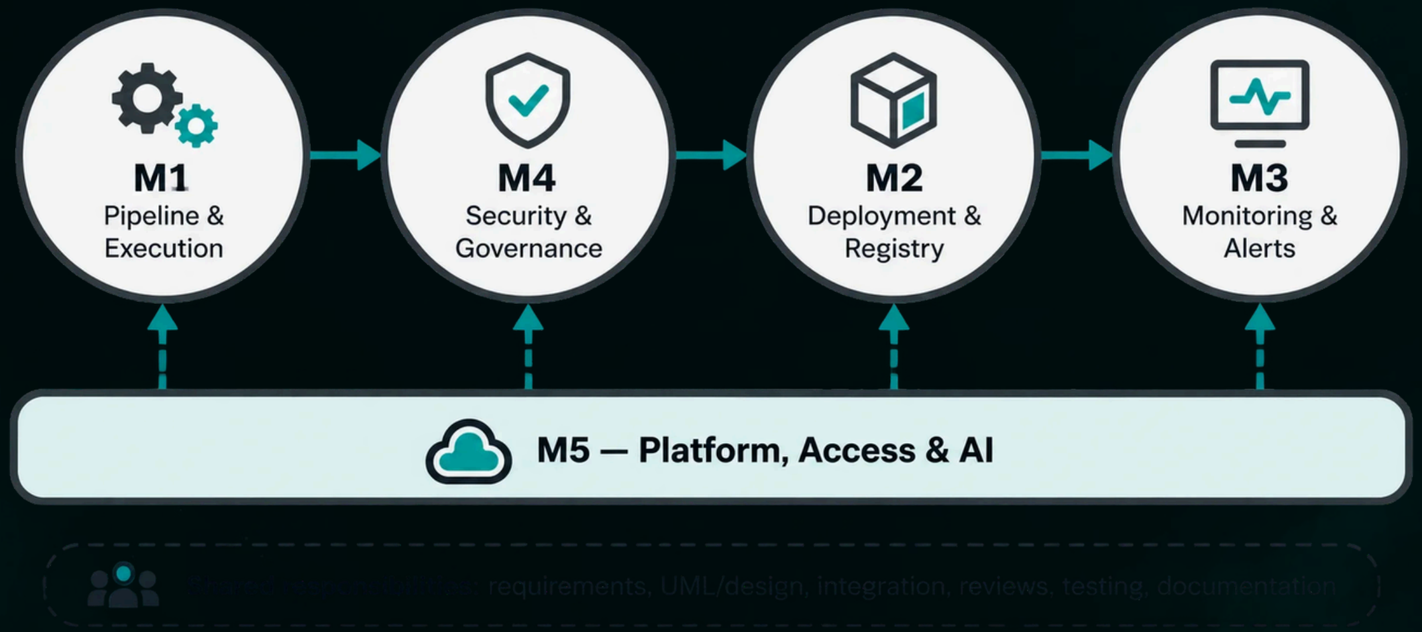

Vercel


Figma

# PROJECT TIMELINE



# INDIVIDUAL CONTRIBUTION



# W.W.A.S.T FERNANDO

## Pipeline & Execution

Frontend

- GitHub repository connection
- Pipeline setup and configuration
- Pipeline dashboard & run history
- Live pipeline status and execution logs
- Build and test result views

Backend

- GitHub webhook integration
- Pipeline run creation and lifecycle
- Queue & background worker processing
- Exact source-revision checkout
- Isolated build/test execution
- Compilation and automated testing
- Pipeline status, logs and failure handling

Key focus:
GitHub → Pipeline → Build/Test → Results

# J.M.D.S. JAYASIRI

## Blockchain Deployment & Contract Registry

Frontend

- Deployment request workflow
- Network and deployment configuration
- Live transaction status tracking
- Contract Registry
- Contract version and deployment history
- Contract details and deployed addresses

Backend

- Deployment validation and execution
- EVM / RPC integration
- Transaction signing and submission
- Confirmation and receipt tracking
- Deployment failure handling
- Contract registration
- Version and deployment-history management

Key focus:
Approved Release → Deployment → Transaction → Contract Registry

# J.B.A.I.N. JAYAWICKRAMA

## Monitoring, Events & Alerts

Frontend

- Contract monitoring dashboard
- Live contract activity feed
- Event and transaction history
- Alert-rule configuration
- Alert status and history
- Runtime analytics / monitoring views

Backend

- Blockchain event and log retrieval
- ABI-based event decoding
- Runtime event storage
- RPC / WebSocket monitoring
- Duplicate-event handling
- Alert-rule evaluation
- Alert creation and runtime-status processing

Key focus:
Deployed Contract → Events → Monitoring → Alerts

# L.L.K.N. RATHNASIRI

## Security & Governance

Frontend

- Security findings dashboard
- Vulnerability detail views
- Release-policy configuration
- Policy result / pass-block views
- Reviewer approval dashboard
- Approval / rejection workflow
- Governance audit history

Backend

- Security-analyzer integration
- Security finding normalization
- Release-policy evaluation
- Deployment eligibility gates
- Approval / rejection workflow
- Reviewer authorization checks
- Governance and audit records

Key focus:

Security Scan → Policy → Approval → Deployment Gate

# F.F. FAZREEN

## Platform, Access & AI Assistant

Frontend

- Login and registration
- Main ChainOps dashboard
- Workspace / organization management
- Project and environment management
- User and role management & Notification centre
- AI Assistant interface

Backend

- Authentication and session management
- Workspace / organization management
- Project and environment APIs
- Role-Based Access Control
- Shared authorization across modules
- Notification processing
- Restricted read-only AI Assistant
- Shared platform integration

Key focus:

User → Platform Access → RBAC → Shared Services → AI Assistant

# CONCLUSION

ChainOps unifies smart contract development, security, deployment, and monitoring in one platform.
It simplifies complex workflows through automation, governance, and real-time visibility.
Our goal is to make smart contract operations more secure, efficient, and traceable.

# Q & A

# THANK YOU !

